SANTOSH KUMAR GUPTA



CONTACT

Address: Hyderabad, Telangana

Email: 23semd04@uohyd.ac.in
WWW: https://www.linkedin.com/in/santosh-kumar-gupta-06b370399/
WWW: https://orcid.org/0009-0009-3476-896X
WWW: Bold Profile

SKILLS

- Basic Python
- Prompt Engineering
- ChatGPT for Educational Applications
- Research methodology
- Qualitative and quantitative educational research

REFERENCES

Prof. J. V., Madhusudan
Head and Professor
University of Hyderabad, Hyderabad, India
madhusudanjv@uohyd.ac.in

PROJECTS

- M.Ed. Dissertation: Relationship Between Critical Thinking and Achievement in Physics
- Ph.D. Research: A Study on Automated Formative Assessment Integrated Multidimensional Learners Data

COLLABORATIVE ENGAGEMENTS

Collaborating Researcher, AI in Education Unit (AIEOU)
An interdisciplinary research hub focused on Artificial Intelligence in Education, University of Oxford.
(Since 2025 – Ongoing)
Website: https://aieou.web.ox.ac.uk
Student Member, EdTech Society (ETSociety)
Focus: A professional body bringing together education, technology and community
Membership No.: STU-000870
Website: https://etsociety.org

LANGUAGES

Hindi (Fluent), English (Fluent)

JOURNAL ARTICLES

Published

Gupta, S. K., Kumar, A., & Madhusudan, J. V. (2026). भारतीय दार्शनिक परिप्रेक्ष्य में पूर्व-प्राथमिक शिक्षा [Pre-primary education in the Indian philosophical perspective]. In Bharatiya Bodh: Reviving Indian knowledge system for modern education. https://zenodo.org/records/19614645 Indu Book Services.

Gupta, S. K., & Madhusudan, J. V. (2025). Evaluating large language models for automated assessment and feedback in physics education. Journal of Education Technology, 9(4), 770–781. https://doi.org/10.23887/jet.v9i4.105174

Sharma, S., & Gupta, S. K. (2023). पूर्णांकों को हमने कैसे समझा?. भारतीय आधुनिक शिक्षा,

NCERT, 44(2). https://www.researchgate.net/publication/391524933_purnankom_ko_hamane_kaise_samajha

Gupta, S. K., & Madhusudan, J. V. (2025). शैक्षिक मूल्याँकन में प्रॉम्प्टिंग तकनीकों की भूमिका: शिक्षकों के लिए एक अवधारणात्मक रूपरेखा [The role of prompting techniques in educational assessment: A conceptual framework for teachers]. भारतीय आधुनिक शिक्षा [Bharatiya Aadhunik Shiksha]. National Council of Educational Research and Training. [Accepted 13 August 2026; October 2025 issue]

Forthcoming / In Press

Gupta, S. K., & Madhusudan, J. V. (in press). Adoption of teachers' strategies to address the English reading and writing skills among early graders. Journal of International Students. [Publish in December 2026]

Gupta, S. K., & Madhusudan, J. V. (in press). Knowing the learner matters: Teachers' formative assessment practices and acceptance of automation. In Proceedings of the International Conference on Technology 4 Education (T4E 2026). Indian Institute of Technology Jodhpur. [Conference: 7–9 December 2026]

Gupta, S. K., & Madhusudan, J. V. (in press). Beyond the score: Teachers' use of multidimensional learner-related data in formative assessment and feedback. In Proceedings of epiSTEME-11. Homi Bhabha Centre for Science Education, Tata Institute of Fundamental Research, Mumbai. [Conference: 5–8 January 2027]

Under Review

Gupta, S. K., & Madhusudan, J. V. (2026). स्कूली शिक्षा में स्वचलित रचनात्मक मूल्यांकन के लिए एक रूपरेखा [A framework for automated formative assessment in school education] [Manuscript submitted for publication]. Submitted to परिप्रेक्ष्य [Paripreksya], National Institute of Educational Planning and Administration. (Expected Jan 2027)

CONFERENCE PRESENTATIONS AND PUBLICATIONS

- **Gupta, S. K. and Madhusudan J. V. (2024, December).** *लार्ज लैंग्वेज मॉडल्स में प्रॉम्प्टिंग तकनीक की सहायता से भौतिकी समस्याओं का अध्ययन.* तकनीकी हिंदी संगोष्ठी – IIT Jodhpur. [Oral Presentation] (Published in Conference Proceedings) https://www.researchgate.net/publication/386469138
- **Gupta, S. K. and Madhusudan J. V. (2025, February).** *Redefining Teacher Roles and Personalized Learning: Educational Data Mining-Based Automated Assessment in School Education.* IOE-UOH National Conference on Educational Technology Post-NEP 2020, University of Hyderabad. [Oral Presentation]
- **Gupta, S. K. and Madhusudan J. V. (2025, December 5–7). "***Can ChatGPT Automate Qualitative Thematic Analysis? A Comparative Study of Input Design and Structured Prompting Strategies."* International Conference on Technology 4 Education (T4E 2025), IIT Madras, Chennai. Organized by EdTech Society. (Accepted for publication in conference proceedings) https://doi.org/10.2139/ssrn.6547580 **[International Conference]**
- **Gupta, S. K. and Madhusudan J. V. (2025, December 22–23). "***Exploring Technological Interventions and Instructional Strategies for Enhancing English Reading and Writing Skills."* National Seminar "Bharatiya Bhasha Parivar", KMICS, Hyderabad. Sponsored by Bharatiya Bhasha Samiti, MoE, Govt. of India.

- **Gupta, S. K. and Madhusudan J. V. (2026, January 5–6). "***प्राथमिक स्तर के विद्यार्थियों में अंग्रेजी भाषा के पठन एवं लेखन कौशल की वर्तमान स्थिति का अध्ययन."* अभ्युदय-3: तकनीकी हिंदी संगोष्ठी 2025, IIT Indore. **[Best Paper Award – Protsahan Puraskar]**
- **Gupta, S. K. and Madhusudan J. V. (2026, February 5–6). "***Teachers' Instructional Strategies for Enhancing English Reading and Writing Skills and Addressing Gender Disparities among Early Graders."* International Conference on NEP-2020: Future Pathways for Teacher Education (Hybrid Mode), University of Hyderabad in collaboration with MANUU, Hyderabad. **[International Conference]**

PERSONAL SUMMARY

Doctoral researcher (UGC-JRF) in Education at the University of Hyderabad, working at the intersection of educational technology and formative assessment. My doctoral study design and evaluates automated formative assessment for school education, drawing on a Physics background (M.Sc.) alongside B.Ed. and M.Ed. training. I collaborate with the AI in Education Unit at the University of Oxford, and publish and present in both English and Hindi across national and international forums.

RESEARCH INTERESTS

- Educational Technology
- Automated Formative Assessment in Education
- Science Education
- Educational Data Mining
- Use of LLMs and Prompting in Learning Environments
- Educational Assessment and curriculum Design in Education

EXPERIENCE

Researcher, 08/2023 - Current
Santosh Kumar Gupta - Hyderabad **,** India

- Gathered relevant data for research projects through comprehensive literature reviews.
- Collaborated with multi-disciplinary teams to design experiments and analyze results.
- Developed research methodologies and frameworks in partnership with team members.
- Participated in meetings to update stakeholders on project progress and findings.
- Attended industry conferences to remain informed on emerging theories and developments.

EDUCATION

Ph.D., Education, Expected in 08/2028
University of Hyderabad - Hyderabad, India
Pursuing

M.Ed., Education, 07/2023
University of Allahabad - Allahabad, India

B.Ed., Education, 07/2021
Ewing Christian College, University of Allahabad - Allahabad, India

M.Sc., Physics, 07/2019
University of Allahabad - Allahabad, India

B.Sc., Physics, Chemistry, and Mathematics, 07/2017
University of Allahabad - Allahabad, India

CERTIFICATIONS, FDPS, WORKSHOPS & TRAININGS

- **Completed Prompt Engineering for ChatGPT**
  Organized by: Coursera (Vanderbilt University)
  Date: June 2025
- **Completed Innovative Teaching with ChatGPT**
  Organized by: Coursera (Vanderbilt University)
  Date: June 2025
- **Completed Full Stack Data Analytics V2**
  Organized by: iNeuron.ai
  Date: 17 January 2025
- **Participated in a Three-Part Series of the Web of Science Certification Program 2025**
  Organized by: Web of Science
  Certificate No.: 577796
- **Participated in Research-Based Curriculum Symposium**
  Organized by: Homi Bhabha Centre for Science Education (HBCSE-TIFR), Mumbai
  Date: 2–4 September 2024
- **Completed One Week Online Faculty Development Programme on "Pathways to Progress: AI Driven Education for Viksit Bharat@2047"**
  Organized by: Delhi Teachers University (DTU)
  Date: 10–16 June 2024
- **Completed NEP 2020 Orientation & Sensitization Programme**
  Organized by: Central University of Gujarat under MM-TTP (UGC)
  Date: 12–22 April 2024
- **Indian Knowledge System (IKS): Concepts and Applications in Science**
  Organized by: Indian Institute of Management Bangalore (IIMB)
  Date: 2024
- **Participated in 8th Community Network Xchange (CNX) Asia-Pacific and 6th Digital Citizen Summit (DCS)**
  Theme: Algorithms, AI and Accountability
  Organized by: Digital Empowerment Foundation
  Date: 2024
- **Completed Digital Pedagogy and Assessment**
  Organized by: SWAYAM
  Date: 2023
- **Attended workshop on "Recent Trends in Research and Communication Skills"**
  Organized by: Nehru Gram Bharti University, Prayagraj, Uttar Pradesh, India
  Date: 12 January 2023
- **Completed Course on Computer Concepts (CCC)**
  Organized by: National Institute of Electronics and Information Technology (NIELIT), Ministry of Electronics and Information Technology, Government of India
  Date: October 2021
- **Attended Yusuf Hamied Inspirational Science Programme (Teacher Training Workshops)**
  Organized by: Royal Society of Chemistry (RSC), United Kingdom – held at Ewing Christian College, Allahabad
  Workshops: Moving Towards Active Learning; Chemical Reactions and Equations; The Particle Nature of Matter

Date: 13–14 September 2019

- **Participated in One Day National Seminar on New Materials and Applications (NSNMA-2018)**
  Organized by: Materials Research Society of India (MRSI) – Allahabad Chapter, University of Allahabad
  Contribution: Poster Presentation
  Date: 13 October 2018

NATIONAL AND STATE-LEVEL QUALIFICATIONS AND AWARDS

- UGC JRF (Education), June 2023
- UGC NET (Education), June 2022, and December 2022
- CTET in Science and Mathematics, January 2021, and December 2021
- UPTET in science and mathematics, 2021
- Rajasthan Teachers' Eligibility Test, 2021

EXTRACURRICULAR ACHIEVEMENTS

- **NCC 'A' Certificate**
  Unit: 65 UP Battalion NCC, Faizabad – NCC Directorate, Uttar Pradesh
  Authority: Ministry of Defence, Government of India
  Date: 2011
- **Represented University of Allahabad – North Zone Inter-University Kabaddi Tournament**
  Venue: Guru Gobind Singh Indraprastha University, New Delhi
  Session: 2016–17 – Certified by the Sports Board, University of Allahabad
- **Represented University of Allahabad – North Zone Inter-University Kabaddi Tournament**
  Venue: Chaudhary Charan Singh University, Meerut
  Session: 2014–15 – Certified by the Sports Board, University of Allahabad
- **Bharat Scouts and Guides – Special Introductory Course (B.Ed./D.El.Ed.)**
  Organized by: Uttar Pradesh Bharat Scouts and Guides, Pradeshik Pradhan Kendra, Lucknow
  Venue: Ewing Christian College, Prayagraj

LANGUAGES

English: C1 Advanced (C1)

Hindi: Native Native